
Models Know, But Don't Always Say: A Guided Reproduction of Two Results on Hidden Knowledge in Language Models

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Abstract

A language model's output need not reflect what it internally represents. We reproduce two published results that make this gap concrete. First, following Marks & Tegmark (COLM 2024), we show that LLaMA-2-13B represents the truth of simple factual statements as a linearly accessible direction: true and false statements separate into two clusters under PCA, and linear probes trained on one dataset of statements generalize to others. Second, following Cywiński et al. (2025), we study *Taboo* models—Gemma-2-9B models fine-tuned to use a secret word downstream while never uttering it—and read the secret out of intermediate layers with the logit lens. All results are reproductions of published work with public code and models, performed with inference-only compute. The running theme: *the model knows, but does not necessarily say*—and simple tools from an introductory AI course suffice to see it.

1 Introduction

When a language model asserts a falsehood fluently, does any part of it “know” better? And when a model is explicitly trained to hide a piece of knowledge, does the knowledge stay hidden on the inside? Both questions ask about the gap between what a model *says* and what it *represents*—a gap with direct consequences for AI safety, where monitors that read only a model's text output can miss what the model is actually computing.

This report reproduces two published answers, chosen because they are striking, complementary, and achievable with the tools of an introductory AI course (PCA, logistic regression, linear discriminant analysis):

- **Reproduction A (main).** Marks & Tegmark [Marks and Tegmark, 2024] showed that LLaMA-2 models represent the truth of simple factual statements along linear directions: true and false statements cluster separately under PCA, probes generalize across datasets, and adding the “truth direction” to activations causally flips the model's answers.
- **Reproduction B (secondary).** Cywiński et al. [Cywiński et al., 2025] released *Taboo* models that know a secret word, use it, but never say it. White-box methods (logit lens on intermediate layers) recover the secret far more often than black-box prompting.

We re-run both pipelines end-to-end—activation extraction, probing, and interventions for A; logit-lens extraction for B—on the authors' public code and models, and report where our numbers match the originals. Everything here is a reproduction: each result is labeled with the figure or table of the original paper it corresponds to.

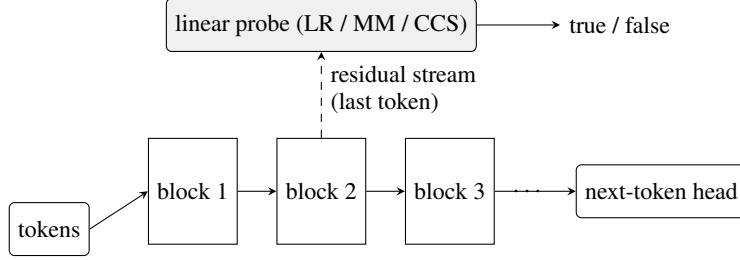


Figure 1: **Reading the residual stream.** After each transformer block, every token position carries a d -dimensional vector. A linear probe trained on these vectors tests whether a property (e.g. truth) is linearly represented; the logit lens instead decodes intermediate states directly into vocabulary space.

2 Background: reading activations

A decoder-only transformer [Touvron et al., 2023, Gemma Team, 2024] maintains, after each block, a *residual stream*: one d -dimensional vector per token position ($d = 5120$ for LLaMA-2-13B). If a property of the input (e.g. truthfulness) is linearly decodable from these vectors, we say the model *linearly represents* it [Alain and Bengio, 2017].

Probes. We use the three probe families of Marks and Tegmark [2024], all bias-free on mean-centered activations: (i) **LR**, logistic regression trained with AdamW; (ii) **MM** (mass-mean), the closed-form mean-difference direction $\theta = \mu^+ - \mu^-$, whose covariance-whitened variant is Fisher’s linear discriminant; (iii) **CCS** [Burns et al., 2023], an unsupervised contrastive probe.

Logit lens. The logit lens [nostalgebraist, 2020] applies the model’s final layer norm and unembedding to intermediate residual streams, reading off which vocabulary tokens the intermediate state already “prefers”. We use it to ask, layer by layer: *where in the network does the hidden knowledge become visible?*

3 Reproduction A: the geometry of truth

3.1 Setup

We follow Marks and Tegmark [2024] (arXiv:2310.06824v3; COLM 2024) and their public repository. **Model:** LLaMA-2-13B [Touvron et al., 2023], bfloat16, inference only. **Datasets:** the repository’s true/false statement datasets—e.g. *cities* (“The city of c is in C .”, 1,496 statements), *neg_cities* (negated variant), *sp_en_trans* (Spanish–English translations), and eight others with fixed templates and known ground truth. **Protocol:** raw declarative statements (no QA formatting, no few-shot priming); we extract the residual stream at the *final token* (the sentence-ending period) at 12 layers spanning the model, with per-dataset mean centering; probes are trained on a random 80/20 split (seed 0). Our activation extractor is a plain-transformers reimplementation that writes the exact file layout of the original repository (the upstream script depends on an outdated `nnsight` API); the probe training code is the authors’ own.

3.2 Result 1: true and false statements separate under PCA

Figure 2 shows the headline phenomenon: without ever seeing labels, PCA on mid-layer activations separates true from false statements into two clusters. Truth is therefore *linearly accessible* in the representation.

3.3 Result 2: probes generalize across datasets (*in progress*)

We train LR, MM, and CCS probes on one (combination of) dataset(s) and evaluate on all eleven (Fig. 3), reproducing the generalization matrix of Fig. 5 of the original paper. Expected landmarks from the original: probes trained on `larger_than+smaller_than` exceed 95% accuracy

Table 1: True/false statement datasets of Reproduction A [Marks and Tegmark, 2024]. The first eight are curated with fixed templates; the last three are uncurated. All have known ground-truth labels.

Dataset	Statement template	n
<code>cities</code>	The city of c is in C .	1,496
<code>neg_cities</code>	The city of c is not in C .	1,496
<code>sp_en_trans</code>	The Spanish word ‘ x ’ means ‘ y ’.	354
<code>neg_sp_en_trans</code>	... does not mean ...	354
<code>larger_than</code>	x is larger than y .	1,980
<code>smaller_than</code>	x is smaller than y .	1,980
<code>cities_cities_conj</code>	It is the case both that s_1 and that s_2 .	1,498
<code>cities_cities_disj</code>	It is the case either that s_1 or that s_2 .	1,498
<code>companies_true_false</code>	company/industry claims [Azaria and Mitchell, 2023]	1,200
<code>common_claim_true_false</code>	natural claims (CommonClaim)	4,450
<code>counterfact_true_false</code>	counterfactual statements (Counterfact)	31,964

Figure pending: PCA of last-token activations (LLaMA-2-13B, layer 14), colored by ground-truth label. Activations are being extracted on the cluster; this panel reproduces the style of Fig. 1 of the original paper.

Figure 2: **Two clusters without any labels.** Each point is one statement’s last-token activation projected onto the top two principal components (PCA is unsupervised). True and false statements separate geometrically. Reproduces the Fig. 1 analysis of Marks and Tegmark [2024].

on `sp_en_trans`; probes trained on `cities` alone fail on `neg_cities` (the two truth directions are nearly orthogonal); adding negated data to training fixes the failure.

3.4 Result 3: pushing along the truth direction flips answers (*in progress*)

Following §6 and Table 2 of the original (reproduced in Table 2): normalize the probe direction θ so that adding it moves the mean false statement onto the mean true one from the probe’s point of view, add it to the residual stream at layers 8–14 at the final tokens, and measure the normalized indirect effect (NIE) on $P(\text{TRUE}) - P(\text{FALSE})$ under a few-shot TRUE/FALSE prompt. The original finds MM directions strongly causal on 13B (NIE up to 0.90) while LR directions are far weaker (0.13–0.19) despite equal classification accuracy—*classification accuracy does not imply causal use*.

4 Reproduction B: reading a secret the model never says

(*In progress; scheduled after the midterm presentation.*) Cywiński et al. [2025] trained *Taboo* models: LoRA adapters on `gemma-2-9b-it` [Gemma Team, 2024] that play a taboo game—they know a secret word (e.g. *gold, moon, cloud*), use it to answer hint questions, and never utter it. We apply the logit lens (Fig. 4) to the residual stream at every layer while the model answers hint prompts, and record the rank of the secret word in each layer’s next-token distribution (Fig. 5). Table 3 lists

Figure pending: train-on-rows / test-on-columns generalization matrix for LR, MM, and CCS probes (LLaMA-2-13B, layer 14).

Figure 3: **Do truth probes generalize?** Each row trains on one (combination of) dataset(s); each column tests on another. Reproduces the Fig. 5 analysis of Marks and Tegmark [2024].

Table 2: Causal-intervention targets from Marks and Tegmark [2024] (their Table 2, LLaMA-2-13B): normalized indirect effect (NIE; 0 = no effect, 1 = full flip) of adding the probe direction to the residual stream. Note the gap: LR and MM classify equally well, but MM directions are far more causal.

Probe direction (13B)	NIE false→true	NIE true→false
LR, trained on cities	0.13	0.19
MM, trained on cities	0.77	0.90
MM, trained on cities+neg_cities	0.85	0.97
CCS, trained on cities+neg_cities	0.31	0.73

the original results we compare against: white-box access lifts an auditor’s top-1 accuracy on Taboo from 42% (text only) to 84% (logit-lens tokens), while the base model without the secret shows no such signal (19%); black-box attacks on Taboo stay below 8%—the secret is hidden from the output, but not from the activations.

5 Discussion and limitations

Everything in this report is a reproduction of published work; our contribution is pedagogical—showing that frontier interpretability results are reachable with course-level tools and inference-only compute. The statements are simple and template-based; truth geometry is richer than a single direction (negated and comparative datasets have near-orthogonal, even antipodal, directions at this scale); and all methods here require white-box access to weights, so they do not transfer to closed API models.

Acknowledgments and Disclosure of Funding

This report reproduces published results: Reproduction A follows Marks and Tegmark [2024] (code: <https://github.com/saprmarks/geometry-of-truth>); Reproduction B follows Cywiński et al. [2025] (code and models: <https://github.com/cywinski/eliciting-secret-knowledge>). All original credit belongs to those authors; any errors in the reproduction are ours.

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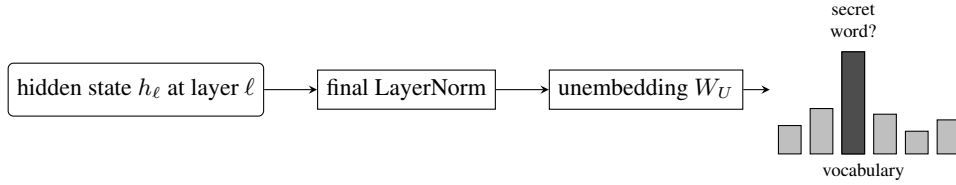


Figure 4: **Logit lens.** Decoding an intermediate hidden state with the model’s own final layer norm and unembedding reveals which tokens the layer “prefers”. For Taboo models, the secret word rises to the top ranks at middle layers even though the output never contains it.

Table 3: Original results from Cywiński et al. [2025] (their Fig. 5–6 and METRICS.md) that we compare against: auditor top-1 accuracy on the Taboo secret word.

Elicitation method	auditor top-1 (%)
Text output only, standard prompts (I/O baseline)	42.1
+ logit-lens tokens (white-box)	84.0
+ activation-similarity tokens (white-box)	85.1
Logit lens on the <i>base</i> model (control)	19.4
Best black-box attack, adversarial prompts	7.2

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*Figure pending: layer $\times$ secret-word rank heatmap (logit lens on
taboo-gold/moon/cloud).*

Figure 5: **Where the secret lives.** Rank of the secret word in each layer’s logit-lens distribution while the model answers hint prompts.